# MATHS AA SL · Rates of change and accumulation

Question booklet · 35 minutes · 28 marks

Answer all questions. A calculator is permitted. Show your reasoning. Give exact answers or round to three significant figures unless directed otherwise. Write on separate paper.

## 1. Zooming in on a filling tank [7]

A sensor models water depth by $d(t) = 12 + 0.8t + 0.06t^2$ centimetres, where $t$ is in seconds. A technician uses the interval from $t = 5$ to $t = 5 + h$ to estimate the rate at $t = 5$.

**(a)** Calculate the average rate for $h = 2$. Give units. **[2]**

**(b)** Show that the average rate for any non-zero admissible $h$ is $1.4 + 0.06h$. Hence find the instantaneous rate. **[3]**

**(c)** For positive $h$, find the largest interval length that gives an estimate within $0.015$ cm/s of the instantaneous rate. **[2]**

## 2. Matching the speed at a join [7]

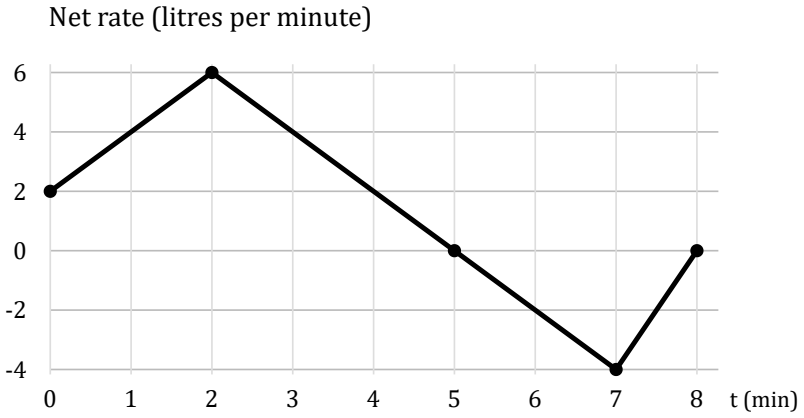
The displacement of a trolley, in metres, is $t^2 + 2t$ for $0 \le t \le 2$ and $a + b(t - 2) - (t - 2)^2$ for $2 < t \le 5$. Time is measured in seconds.

**(a)** Find $a$ and $b$ so that displacement and velocity are continuous at 2 seconds. **[4]**

**(b)** With those values find the acceleration immediately before and after the join. Does continuous velocity guarantee continuous acceleration here? **[3]**

## 3. When a tank is fullest [8]

A tank contains 20 litres initially. Its net inflow rate varies linearly between the points (0, 2), (2, 6), (5, 0), (7, −4) and (8, 0) shown below. Coordinates are (minutes, litres per minute).




**(a)** Find the amount in the tank at 5 minutes. **[3]**

**(b)** Find the amount at 8 minutes. Distinguish net change from total absolute flow. **[3]**

**(c)** State when the tank is fullest and justify using the graph. **[2]**

## 4. A calibration limit [6]

During a calibration the output is $S(t) = 30 + 12t - 3t^2$ for $0 \le t \le 4$. Engineers accept the model while its output is increasing and its instantaneous rate is at least 3 units per second.

**(a)** Find the times for which both conditions hold. **[3]**

**(b)** A straight-line estimate from $t = 1$ to $t = 2$ has slope 3. Explain why the whole interval cannot be accepted. **[3]**

*End of question booklet · 28 marks*